

E-COMMERCE SALES PERFORMANCE DASHBOARD REPORT

1. Project Overview

Project Title: E-Commerce Sales Performance Dashboard

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Tools Used: Excel, PostgreSQL, Power BI, SQL, DAX

2. Abstract / Executive Summary

This project presents a complete end-to-end data analytics solution for analyzing e-commerce sales performance. I designed and implemented a PostgreSQL database to manage large-scale transactional data and optimized it using SQL views and indexing techniques. The processed data was then integrated with Power BI to build a highly interactive and performance-efficient dashboard.

The dashboard provides actionable insights into revenue trends, customer behavior, delivery performance, product performance, and seller contributions. A hybrid approach using Import and DirectQuery modes ensures both high performance and near real-time analytics.

3. Introduction

In this project, I developed a scalable business intelligence solution capable of handling large datasets efficiently. The system demonstrates real-world industry practices such as database optimization, star schema modeling, and interactive dashboard creation.

The goal was to transform raw e-commerce data into meaningful insights that support business decision-making.

4. Problem Statement

Handling large-scale e-commerce datasets in Power BI using DirectQuery leads to performance issues due to complex joins and high data volume. Queries become slow, and dashboard responsiveness is affected.

The challenge was to design an optimized system that ensures fast performance, efficient data modeling, and real-time insights without overloading the BI layer.

5. Objectives of the Project

- Build a PostgreSQL database for e-commerce data
- Design an optimized star schema
- Import large datasets efficiently using chunking
- Create SQL views to reduce query complexity

- Apply indexing for faster performance
 - Integrate PostgreSQL with Power BI
 - Develop DAX measures for KPIs
 - Build an interactive multi-page dashboard
-

6. Scope of the Project

This project includes data ingestion, transformation, modeling, and visualization of e-commerce sales data. It covers customers, orders, products, sellers, payments, reviews, and delivery logistics.

Advanced machine learning is not included but can be implemented in future extensions.

7. Dataset Description

- Source: Olist Brazil E-Commerce Dataset (Kaggle) : [LINK](#)
 - **~1 LAKH** customers and multiple transactional tables
 - Includes orders, products, sellers, payments, reviews, and geolocation
 - Large datasets handled using chunk-based import
 - Stored in PostgreSQL tables
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8. Tools and Technologies Used

- Excel – Data Cleaning
 - PostgreSQL – Database Management
 - pgAdmin – Query execution
 - Power BI – Dashboard creation
 - SQL – Data transformation
 - DAX – KPI calculations
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9. System Architecture / Workflow Overview

1. CSV datasets collected
2. PostgreSQL database created
3. Data imported using COPY command in chunks (~1 lakh rows)
4. Star schema designed with fact and dimension tables
5. SQL views created for optimized querying

6. Indexes applied on key columns (order_id, customer_id)
 7. Power BI connected using Import + DirectQuery
 8. DAX measures created
 9. Dashboard built with multiple analytical pages
-

10. Data Collection & Ingestion (PostgreSQL)

- Database *olist_db* created
 - Large CSV files split into smaller chunks
 - COPY command used for fast ingestion
 - Drop-create approach used to avoid duplication
 - Foreign key constraints added after import
-

11. Database Design

11.1 ER Diagram

- Star schema with central fact table: *bi_fact_sales*
- Dimension tables: Date, Customers, Products, Sellers

11.2 Table Structure

- Fact table contains transactional metrics (price, delivery dates, order status)
- Dimension tables contain descriptive attributes

11.3 Relationships

- One-to-many relationships from dimensions to fact
 - Primary and foreign keys maintained for integrity
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12. Data Cleaning & Preprocessing

- Removed unnecessary columns
 - Standardized formats and data types
 - Ensured null handling and consistency
 - Verified relationships after data load
-

13. Data Transformation (SQL Operations)

- Created optimized SQL views to reduce data volume

- Selected only relevant columns (~20 from 70+)
 - Simplified joins within views
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14. Data Modeling

- Implemented star schema in Power BI
 - Used *DimDate* table for time intelligence
 - Connected Power BI directly to SQL views
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15. Analytical Views Creation

- Created *bi_fact_sales* view for consolidated analysis
 - Reduced complexity and improved performance in Power BI
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16. Key Business Metrics Defined

Overview Page

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value (AOV)
- On-Time Delivery %
- Average Rating
- Year-over-Year Growth (YoY %)

Delivery & Logistics

- On-Time %
- Late Orders %
- Average Delivery Days

Reviews

- Positive Reviews %
- Negative Reviews %
- Average Review Score

Sellers

- Revenue per Seller

- Total Sellers

Drill-Through Page

- Total Revenue
 - Total Orders
 - AOV
 - Avg Rating
 - On-Time %
-

17. Data Validation & Quality Checks

- Verified row counts after ingestion
 - Cross-checked SQL outputs with Power BI visuals
 - Ensured data consistency across transformations
-

18. Integration with Power BI

- Import mode for dimension tables
 - DirectQuery for fact table (bi_fact_sales)
 - Hybrid model improves performance and data freshness
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19. Dashboard Design & Development

Dashboard Pages Created

- Overview
- Sales Trends
- Category & Products
- Customer Geographics
- Delivery & Logistics
- Reviews
- Payments
- Sellers
- Drill-Through Analysis

Features Implemented

- Drill-through navigation
 - Interactive slicers and filters
 - Dynamic titles using measures
 - Multi-page navigation
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20. Data Visualization Techniques Used

- KPI cards for summary metrics
 - Line charts for sales trends
 - Bar charts for category comparison
 - Maps for geographic analysis
 - Donut charts for distribution
 - Matrix tables for detailed breakdown
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21. Key Insights & Findings

- Majority of orders delivered on time
 - Certain regions show higher delivery delays
 - Positive reviews dominate customer feedback
 - Revenue concentrated in specific product categories
 - Credit card is the most used payment method
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22. Business Recommendations

- Improve logistics in delay-prone regions
 - Focus on high-performing categories
 - Optimize seller performance monitoring
 - Enhance customer experience using review insights
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23. Performance Optimization Techniques

- SQL views to reduce dataset size
- Indexing on order_id, customer_id
- Star schema for efficient joins

- Hybrid connection (Import + DirectQuery)
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24. Challenges Faced

- Handling large datasets exceeding system limits
 - Performance issues in DirectQuery mode
 - Complex joins causing slow queries
 - Errors during data import and transformation
 - Understanding and implementing optimal data modeling strategies
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25. Solutions Implemented

- Split large files into chunks
 - Created optimized SQL views
 - Applied indexing for faster queries
 - Used hybrid data connection approach
 - Improved model design using star schema
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26. Future Enhancements

- Real-time data streaming
 - Sales forecasting models
 - Customer segmentation
 - Advanced anomaly detection
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27. Conclusion

This project demonstrates a complete industry-level data analytics workflow. By combining PostgreSQL and Power BI, I successfully built a high-performance dashboard that transforms raw data into actionable business insights. The project highlights my ability to handle large datasets, optimize performance, and deliver meaningful analytics solutions.

28. References

- Kaggle Olist Dataset : [LINK](#)
- PostgreSQL Documentation
- Power BI Documentation

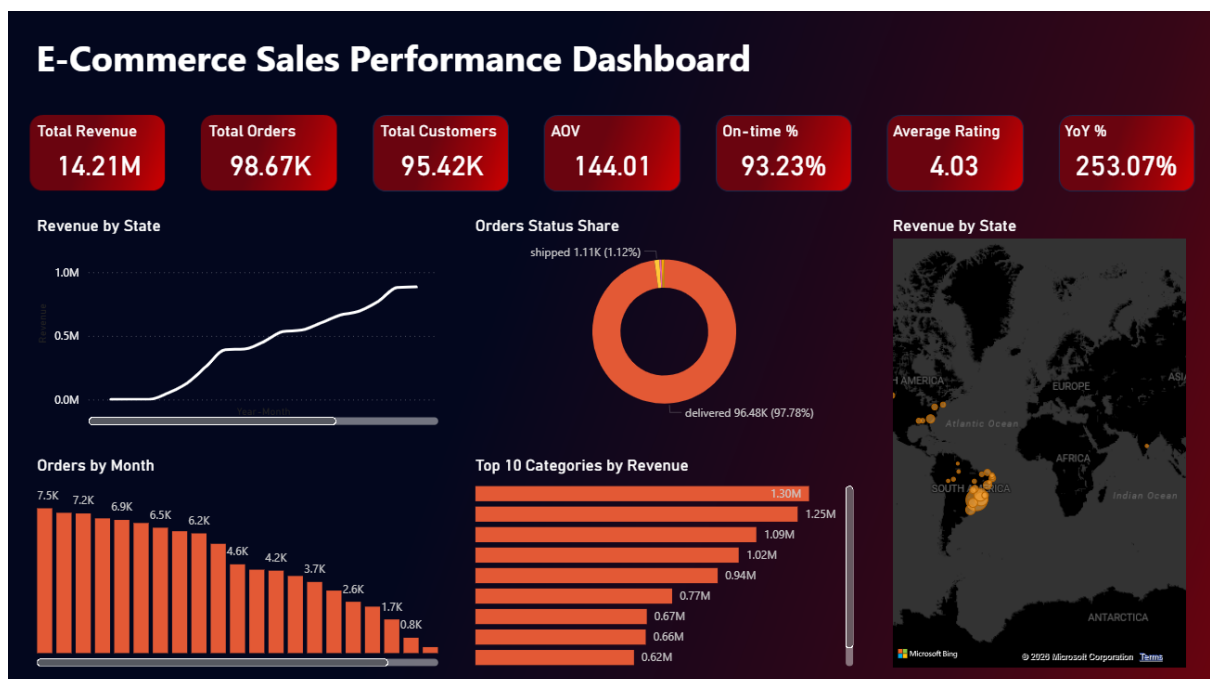
29. Appendix

- SQL scripts (views, indexing)
 - DAX measures
 - Dashboard screenshots
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30. Project links

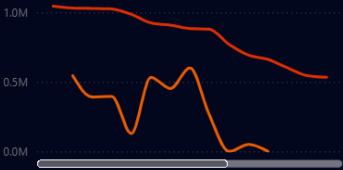
- GitHub Repository: [<https://github.com/mbcgit-hub18/E-Commerce-Data-Analytics-Project>]
- Power BI Dashboard: [Screenshot]

OVERVIEW:



SALES TRENDS::

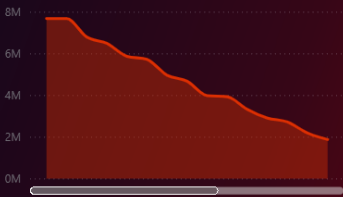
Revenue Vs Last Year



Rolling 30D Revenue



Revenue YTD



Category X year

Year	2016			2017			2018			Total		
category	Revenue	Orders	AOV	Revenue	Orders	AOV	Revenue	Orders	AOV	Revenue	Orders	AOV
agro_industry_and_commerce	65.89	2.00	32.95	1,13,825.97	822.00	138.47	77,064.73	627.00	122.91	1,90,956.59	1451.00	131.60
air_conditioning	1,707.09	5.00	341.42	29,223.18	48.00	608.82	54,618.77	134.00	407.60	83,841.95	182.00	460.67
art				9,454.14	36.00	262.62	15,267.83	166.00	91.97	24,721.97	202.00	122.39
arts_and_craftmanship				151.89	2.00	75.95	1,662.12	21.00	79.15	1,814.01	23.00	78.87
audio	156.99	2.00	78.50	18,579.85	161.00	115.40	33,713.53	187.00	180.29	52,450.37	350.00	149.86
auto	1,833.25	11.00	166.66	2,51,687.77	1429.00	176.13	3,63,231.49	2457.00	147.84	6,16,752.51	3897.00	158.26
baby	1,630.16	11.00	148.20	1,58,314.64	1207.00	131.16	2,74,534.81	1667.00	164.69	4,34,479.61	2885.00	150.60
bed_bath_table	478.99	5.00	95.80	5,30,875.53	4503.00	117.89	5,61,196.50	4909.00	114.32	10,92,551.02	9417.00	116.02
books_general_interest	119.50	1.00	119.50	20,288.19	221.00	91.80	27,789.76	290.00	95.83	48,197.45	512.00	94.14
books_imported				1,123.99	13.00	86.46	3,727.86	40.00	93.20	4,851.85	53.00	91.54
books_technical	267.00	1.00	267.00	5,807.22	43.00	135.05	13,196.69	216.00	61.10	19,270.91	260.00	74.12
cds_dvds_musicals				665.00	11.00	60.45	65.00	1.00	65.00	730.00	12.00	60.83
christmas_supplies				2,861.11	50.00	57.22	5,965.73	78.00	76.48	8,826.84	128.00	68.96
Total	51,220.49	312.00	164.17	64,83,893.86	44579.00	145.45	76,74,135.96	53775.00	142.71	1,42,09,250.31	98666.00	144.01

CUSTOMER GEOGRAPHICS:

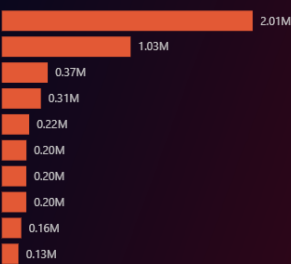
Customer State X Category

category	agro_industry_and_commerce		air_conditioning		art		arts_and_craftmanship		audio		auto		baby	
customer_state	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders	Revenue	Orders
SP	73,864.79	589.00	35,413.05	85.00	23,111.84	117.00	15,162.18	94.00	906.25	15.00	18,071.98	139.00	2,21,559.20	1615.00
RJ	24,624.96	199.00	9,440.84	25.00	16,555.20	60.00	3,202.89	32.00	10,009.24	54.00	68,510.63	413.00	44,264.47	442.64
MG	18,366.42	172.00	13,066.99	14.00	4,505.01	18.00	1,441.54	18.00	488.78	6.00	5,676.89	33.00	75,238.22	471.00
RS	8,415.30	84.00	3,155.93	9.00	1,702.19	12.00	611.44	9.00			2,436.10	16.00	27,414.39	176.00
PR	8,577.23	73.00	4,903.53	12.00	3,706.33	10.00	710.82	9.00			2,871.17	17.00	28,612.81	209.00
BA	7,821.85	46.00	2,257.00	3.00	832.86	6.00	898.95	10.00			1,430.69	14.00	30,887.80	148.00
SC	7,321.55	62.00	1,069.98	3.00	599.60	1.00	422.19	6.00	289.49	1.00	1,822.56	10.00	29,723.77	164.00
DF	2,701.81	32.00			468.00	3.00	565.79	6.00	129.49	1.00	2,048.23	16.00	17,389.04	104.00
Total	1,90,956.59	1451.00	83,841.95	182.00	55,962.38	253.00	24,721.97	202.00	1,814.01	23.00	52,450.37	350.00	6,16,752.51	3897.00

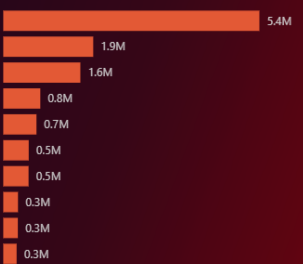
Revenue by State



Top 10 Customer Cities by Revenue



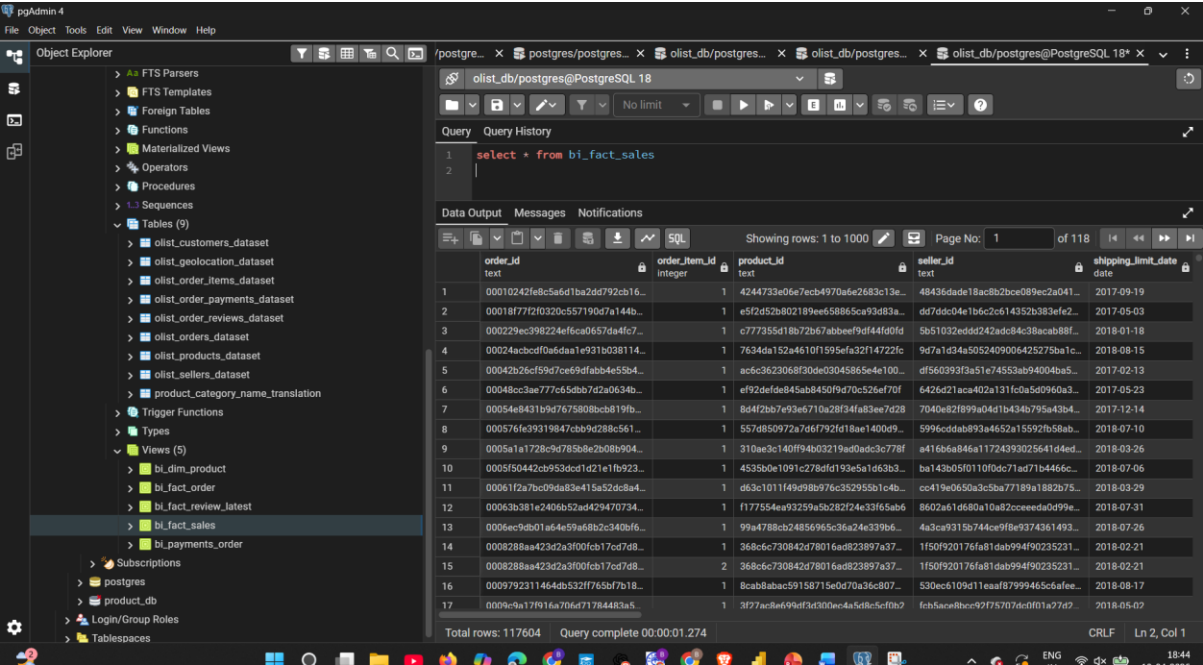
Top 10 Customer States by Revenue



PostgreSQL Tables:

VIEWS:

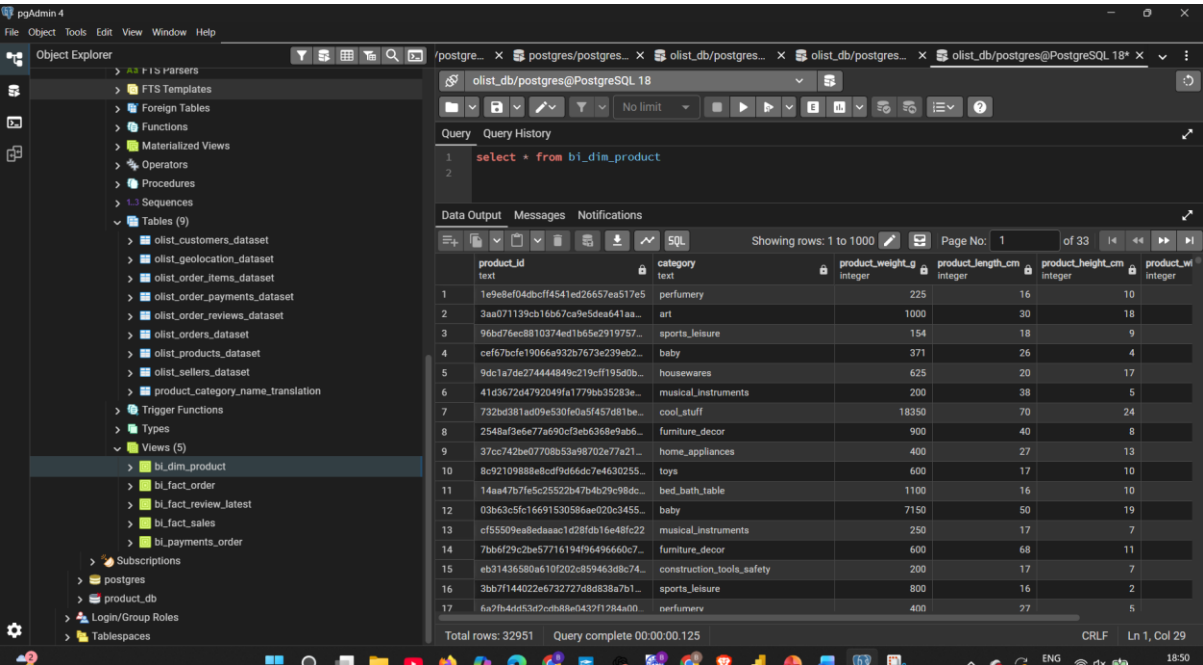
 bi_fact_sales:



The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including tables like `olist_customers_dataset`, `olist_geolocation_dataset`, `olist_order_items_dataset`, `olist_order_payments_dataset`, `olist_order_reviews_dataset`, `olist_orders_dataset`, `olist_products_dataset`, `olist_sellers_dataset`, `product_category_name_translation`, `bi_dim_product`, `bi_fact_order`, `bi_fact_review_latest`, `bi_fact_sales`, and `bi_payments_order`. The main pane shows the query results for the `bi_fact_sales` table. The query is `select * from bi_fact_sales`. The results show columns: `order_id` (text), `order_item_id` (integer), `product_id` (text), `seller_id` (text), and `shipping_limit_date` (date). The table contains 117604 rows.

order_id	order_item_id	product_id	seller_id	shipping_limit_date
00010242fe8c5a6d1ba2dd792cb16...	1	4244733e06e7ecb4970a5e2683c13e...	48436dade18ac8b2bce089ec2a041...	2017-09-19
00018f77f2f0320c557190d7a144b...	1	e5f2d52b802189ee658865ca93d83a...	d87ddcd04e1b6c2c614352b383fe2...	2017-05-03
000229ec398224fe6c0657da4fc7...	1	c777355d18b72b67abbeef9df44f00d...	5b51032eddd242adc84c38acab88f...	2018-01-18
00024acbcd0fa5daa1e931b038114...	1	7634da152a4610f1595efa32f14722fc...	9d7a1d34a5052409006425275ba1c...	2018-08-15
00042b26cf59d7ce69dfabb4e55b4...	1	ac5c3623068f30de03045865e4e100...	df560393f3a51e74533ab94004ba5...	2017-02-13
00048cc3ae777c65bb7d2a0634b...	1	ef92dfe4e845ab8450f9d70c526ef70f...	6426d21aca402a131fc0a5d0960a3...	2017-05-23
00054e8431b9d7675808bcb819fb...	1	8d4f2b67e93ae710a28f34fa83ee7d28...	7040e2f899a04d1b434b795a43b4...	2017-12-14
000576fe39319847cbb9d288c561...	1	557d850972a7d6f792d18ae1400d9...	5996cddad893a4652a15592fb58ab...	2018-07-10
0005a1e1728c9d785b8e2b080904...	1	310ae3c140ff94b03219ad0adc3c778f...	a416b6a846a11724393025641d4ed...	2018-03-26
0005f50442cb953dcfd121e1fb923...	1	453b0e1091c278afd193e5a1d63b3...	ba143b05f0110f0dc71ad71b4466c...	2018-07-06
00061f2a7bc09da83e415a52dc8a4...	1	d63c1011f49d98b976c352955b1c4b...	cc419e0650a3c5ba77189a1882b75...	2018-03-29
00063b381e240b52ad429470734...	1	f177554ea93259a5b282724e33f65ab6...	8602a51d680a10a82ceeeada0d99e...	2018-07-31
0006ec9db01af64e59af68b2c340bf6...	1	99a4788cb248569e5c36a24e339b6...	4a3ca9315b744ce9f8e9374361493...	2018-07-26
0008288aa423d2a3f00fcb17cd7d8...	1	368c6c730842d78016ad823897a37...	1f50f920176fa81dab999490235231...	2018-02-21
0008288aa423d2a3f00fcb17cd7d8...	2	368c6c730842d78016ad823897a37...	1f50f920176fa81dab999490235231...	2018-02-21
0009792311464db332f176bf7b18...	1	8cab8abac59158715e0d70a36c807...	830ec109d11eaaf879994656cafee...	2018-08-17
0009c9a17f916a70f6f71784483a...	1	3d72ac8e4694df3d300dec4a5d8c5cf0b...	fc83ace8b9cc92f7570f0d0f1a72d...	2018-05-07

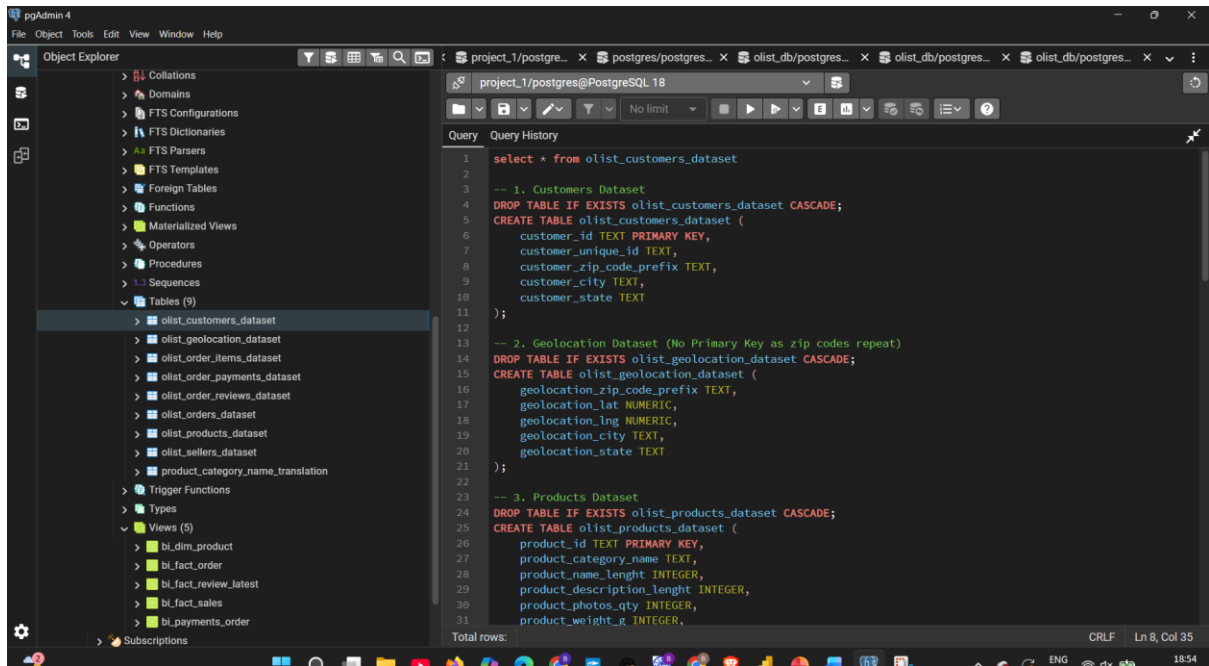
 bi_fact_sales:



The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including tables like `olist_customers_dataset`, `olist_geolocation_dataset`, `olist_order_items_dataset`, `olist_order_payments_dataset`, `olist_order_reviews_dataset`, `olist_orders_dataset`, `olist_products_dataset`, `olist_sellers_dataset`, `product_category_name_translation`, `bi_dim_product`, `bi_fact_order`, `bi_fact_review_latest`, `bi_fact_sales`, and `bi_payments_order`. The main pane shows the query results for the `bi_dim_product` table. The query is `select * from bi_dim_product`. The results show columns: `product_id` (text), `category` (text), `product_weight_g` (integer), `product_length_cm` (integer), `product_height_cm` (integer), and `product_weight` (integer). The table contains 32951 rows.

product_id	category	product_weight_g	product_length_cm	product_height_cm	product_weight
1e9e8ef04dbcf4541ed26657ea517e5	perfumery	225	16	10	
3aa071139cb16b67ca9e5dea641aa...	art	1000	30	18	
96bd76ec8810374ed1b65e2919757...	sports_leisure	154	18	9	
cef67bce19066a932b767e239eb2...	baby	371	26	4	
9dc1a7de27444849c219cf19500b...	housewares	625	20	17	
41d3672d4792049fa1779bb35283e...	musical_instruments	200	38	5	
732bd381ad09e530fe0a5f457d81be...	cool_stuff	18350	70	24	
2548af3e6e77a590cf3eb6368e9ab6...	furniture_decor	900	40	8	
37cc742be07708b53a98702e77a21...	home_appliances	400	27	13	
8c92109888e8cd9d6dc7e4630255...	toys	600	17	10	
14aa47b7fe5c25522b47b4b29c98dc...	bed_bath_table	1100	16	19	
03b63c5fc16691530586ae020c3455...	baby	7150	50	10	
ef5509ea8edaac1d28fdb16e48fc22...	musical_instruments	250	17	7	
7bb6f29c2be577161949646660c7...	furniture_decor	600	68	11	
eb31436580a610f020c859463d8c74...	construction_tools_safety	200	17	7	
3bb7f144022e6732727d8d838a7b1...	sports_leisure	800	16	2	
6a7b4d453d7c8b88a0432f128a40...	perfumery	400	27	5	

Sample SQL View:



-----**THE END**-----